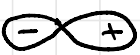
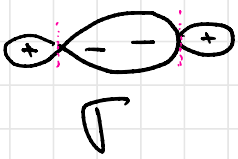
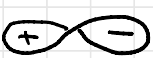
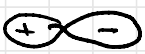
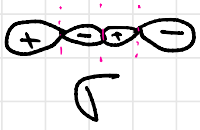
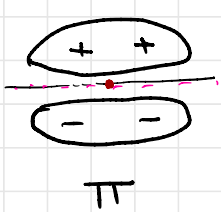
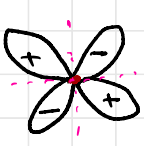
CH1101 : Quantum Chemistry & Spectroscopy

29/09/2026

Lec 14

Chemical bonding : Homonuclear diatomics

AO1	AO2	MO		Symmetry
 1s	 1s	 σ	Bonding	σ_g
 1s	 1s	 σ	Anti-bonding	σ_u
 p_z	 p_z	 σ	Bonding	σ_g
 p_z	 p_z	 σ	Anti-bonding	σ_u
 p_x	 p_x	 π	Bonding	π_u
 p_x	 p_x	 π	Anti-bonding	π_g

https://huggingface.co/spaces/rshampa/pltorb

interactive AO viewer

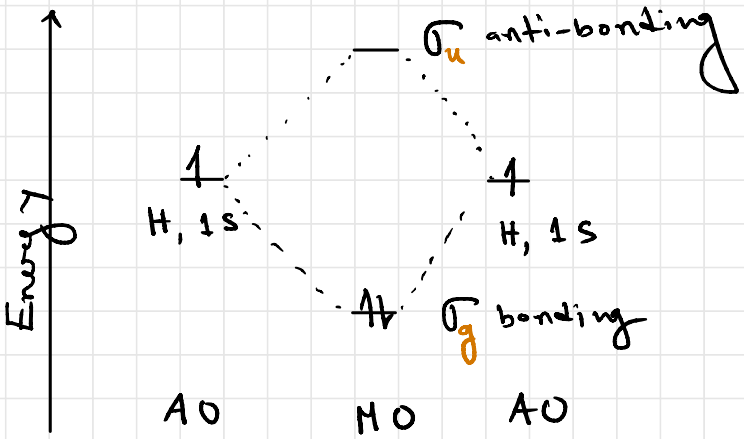
AO : Atomic Orbital

MO : Molecular Orbital

Molecular Orbital Energy Diagram

No. of electrons = 2

H_2



Electronic configuration of H_2 : σ_g^2

$$\left\{\begin{array}{l} BO = \dfrac{2-0}{2} = 1 \\ \text{of} \\ H_2 \end{array}\right.$$

Bond-Order =

$$\frac{1}{2}\left(\text{No. of bonding electrons} - \text{no. of anti-bonding electrons}\right)$$

He_2 No. of electrons = 4

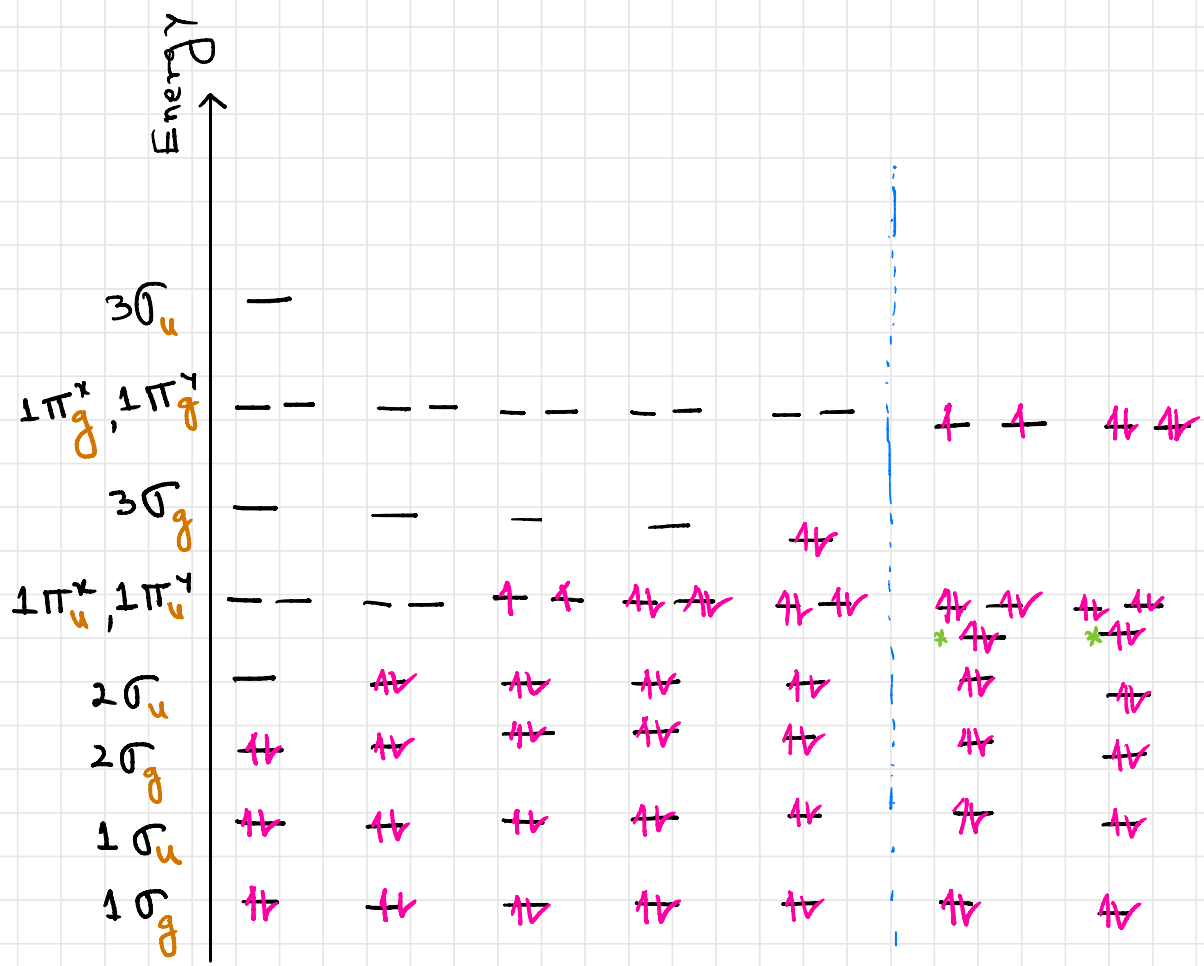
Electronic configuration : $\sigma_g^2 \sigma_u^2$

Bond-Order

$$BO = \frac{2-2}{2} = 0$$

MOT fails to explain the stability of He_2 !

Arrange MOs according to their energies



Molecule	Li_2	Be_2	B_2	C_2	N_2	O_2	F_2
No. of electrons	6	8	10	12	14	16	18
BO	1	0	1	2	3	2	1

* Why lower?
Due to S- P orbital interaction!

Electronic Configuration of O_2 :

$$1\sigma_g^2\ 1\sigma_u^2\ 2\sigma_g^2\ 2\sigma_u^2\ 3\sigma_g^2\ 1\pi_u^{x^2}\ 1\pi_u^{y^2}\ 1\pi_g^{x^1}\ 1\pi_g^{y^1}\ 3\sigma_u$$

$$BO = \frac{10-6}{2} = \frac{4}{2} = 2$$

BO of F_2 = 1

O_2 & F_2 } which one will have higher dissociation energy ?

Ans O_2 as, O_2 has higher BO than F_2

How about BO of O_2^+ ?

O_2^+ : No. of electrons = 15 // $BO = \frac{10-5}{2} = 2.5$

higher the Bond order
- higher bond energy
- higher bond dissociation energy
- higher bond strength
- smaller bond length